

Homo. ① Diff eqn

$$\frac{dy}{dx} = f(x, y) \text{ is Homo.}$$

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Eg. $\frac{dy}{dx} = \frac{2(x+y)}{x}$

$$= 2 + 2\left(\frac{y}{x}\right) \rightarrow \textcircled{1}$$

$$\text{Set } v = \frac{y}{x}$$

$$vx = y$$

$$v + \frac{dv}{dx} x = \frac{dy}{dx} \rightarrow \textcircled{2}$$

$$\textcircled{1}, \textcircled{2} \Rightarrow v + \frac{dv}{dx} x = 2 + 2v$$

$$dx$$

$$\Rightarrow \frac{dv}{dx} x = 2 + v$$

$$\Rightarrow \frac{dv}{2+v} = \frac{dx}{x}$$

$$\Rightarrow \ln|2+v| = \ln|x|$$

$$\Rightarrow |2+v| = |x|$$

$$\Rightarrow |2 + \frac{y}{x}| = \frac{|x|}{x^2}$$

$$\Rightarrow |2x + y| = x$$

$$\Rightarrow 2x^2 + y = \pm x$$

$$\Rightarrow y = \pm x^2 - 2x$$

$$y'' + 2y' + y = 0$$

$$\checkmark x^2 + 2x + 1 \quad Ay'' + By' +$$

$$C=0$$

$$(x+1)^2$$

$$x = -1 \text{ soln}$$

$$-x \quad \text{---} \quad -x e, \quad x e$$

$$GS = C_1 e^{-x} + C_2 x e^{-x}$$

$$A, B \leftarrow \text{Distinct}$$

$$C_1 e^{Ax} + C_2 e^{Bx} \leftarrow$$

$$A \text{ --- Repeated}$$

$$C_1 e^{Ax} + C_2 x e^{Ax} \leftarrow$$

$A \pm iB$ - Complex

$$e^{Ax} \left(c_1 \cos Bx + c_2 \sin Bx \right)$$



$$Ay'' + By' + Cy = 0.$$

$$y'' + 4y' + 3y = 7e^{4x} \rightarrow \textcircled{1}$$

$$x^2 + 4x + 3 = 0 \Rightarrow (x+3)(x+1) = 0$$

$$\Rightarrow \underline{c_1 e^{-3x} + c_2 e^{-x}} \text{ is}$$

a soln of homo. eqn

Particular Solution (PS)

Suppose Ae^{4x} is a PS

$$\text{Sup } y = Ae^{4x}$$

$$y' = 4Ae^{4x}$$

$$y'' = 16Ae^{4x}$$

$$\text{Hence } \textcircled{1} \Rightarrow 16Ae^{4x} + 16Ae^{4x} + 3Ae^{4x} = 7e^{4x}$$

$$\Rightarrow 35A = 7$$

$$\Rightarrow A = \frac{1}{5}$$

Hence

$$GS = \underline{C_1 e^{-3x} + C_2 e^{-x} + \frac{1}{5} e^{4x}}$$

Laplace Transform

Analysis $\xrightarrow{\mathcal{L}}$ Algebra

Time Domain $\xrightarrow{\mathcal{L}}$ Frequency Domain

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$
$$= F(s)$$

$$\mathcal{L}\{1\} = 1/s ; s > 0$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a} ; s > a$$

$$\mathcal{L}\{1\} = \int_0^{\infty} e^{-st} dt$$

$$= \frac{e^{-st}}{-s} \Big|_0^{\infty}$$

$$= 0 - (-1/s)$$

$$= 1/s \quad (s > 0)$$

$$\mathcal{L}\{e^{at}\} = \int_0^{\infty} e^{-st} e^{at} dt$$

$$= \int_0^{\infty} e^{-(s-a)t} dt$$

$$= \frac{e^{-(s-a)t}}{-(s-a)} \Big|_0^{\infty}$$

$$= (0 + 1/s-a); \text{ if } s > a$$

(otherwise $\int$)

doesn't
exist)

$$= \frac{1}{s-a}$$

$$\mathcal{L}\{\sin(at)\} = \int_0^{\infty} e^{-st} \sin(at) dt$$

$$\underline{u = e^{-st}} \quad , \quad \underline{dv = \sin(at)}$$

$$= -\underline{e^{-st} \cos(at)} \Big|_0^{\infty} + \frac{1}{a} \int_0^{\infty} \cos(at) e^{-st} dt$$

$$= (0 + 1/a) + -\frac{s}{a} \mathcal{L}\{\cos(at)\}$$

$$\underline{\mathcal{L}(\cos(at))} \quad (\text{sup } s > 0)$$



①

$$u_{\cos(at)} = e^{-st}, \quad dv =$$

$$\mathcal{L}\{\cos(at)\} = \frac{e^{-st} \sin(at)}{a} \Big|_0^\infty$$

$$- \int_0^\infty \frac{\sin(at)}{a} e^{-st} (-s)$$

$$= (0 - 0) + \frac{s}{a} \mathcal{L}(\sin(at))$$

$$(\textcircled{2} \sup s > 0) \rightarrow$$

$$\textcircled{1}, \textcircled{2} \Rightarrow \underline{\mathcal{L}(\sin(at))}$$

$$= \frac{1}{a} \mathcal{L}(\sin(at)) = \frac{s/a}{s^2/a^2 + 1}$$

$$= 1/a - s^2/a^2 \mathcal{L}(\sin at)$$

$$\Rightarrow \left(1 + \frac{s^2}{a^2}\right) \mathcal{L}(\sin at) = 1/a$$

$$\Rightarrow \mathcal{L}(\sin at) = \frac{a}{s^2 + a^2} \rightarrow \textcircled{3}$$

$$\textcircled{2}, \textcircled{3} \Rightarrow \mathcal{L}(\cos at)$$

$$= \frac{s}{s^2 + a^2}$$

$$\mathcal{L}\{t\} = \int_0^{\infty} e^{-st} t \, dt$$

$$u = t, \quad dv = e^{-st}$$

$$= \left[t \frac{e^{-st}}{-s} \right]_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} \, dt$$

-st

$$= \left(\begin{array}{c} 0 \\ \underline{e} \\ -s \end{array} \right) (0) + \frac{1}{s}$$

(Assume $s > 0$)

$$= \frac{-1}{s^2} (0 - 1)$$

$$= \frac{1}{s^2}$$

$$\mathcal{L}\{f'(t)\} = \int_0^{\infty} e^{-st} f'(t) dt$$

$$u = e^{-st} \quad dv = f'(t)$$

$$= \left[e^{-st} f(t) \right]_0^{\infty}$$

$$- \int_0^{\infty} f(t) e^{-st} (-s) dt$$

$$= (0 - f(0))$$

$$+ s \mathcal{L}(f(t)) \quad (\text{Assume } s > 0)$$

$$\underline{\mathcal{L}\{f'(t)\} = s \mathcal{L}(f(t)) - f(0)}$$

$$\text{Set } f(t) = t^2$$

$$\mathcal{L}\{t^2\} = \frac{\mathcal{L}\{2t\} + f(0)}{s}$$

$$= \frac{2(1/s^2) + f(0)}{s}$$

$$= \frac{2}{s^3}$$

$$\mathcal{L}(t^3) = 3! / s^4$$

$$\mathcal{L}(t^n) = n! / s^{n+1} ; \forall n \in \mathbb{N}_{\geq 0}$$

Chain Rule

$$y = f(u)$$

$$z = g(y)$$

$$\frac{dz}{du} = \frac{dz}{dy} \frac{dy}{du} \quad \angle$$

$$\rightarrow \frac{g(f(u_1)) - g(f(u_2))}{f(u_1) - f(u_2)} \leftarrow$$

$$\rightarrow \frac{f(u_1) - f(u_2)}{u_1 - u_2} \leftarrow$$

$$y = u^2, \quad z = \sin(y)$$

$$\frac{dz}{du} = \frac{\cos(y)}{\cos(u^2)} \cdot \frac{2u}{2u}$$

$$y'' + 4y' + 3y = \sin 3x \rightarrow \textcircled{1}$$

$$y_h = C_1 e^{-3x} + C_2 e^{-x}$$

Sup ^{y_p} " $A \sin 3x + B \cos 3x$
is a soln of $\textcircled{1}$ " $\leftarrow$

$$, y_p = 3A \cos 3x - 3B \sin 3x$$

$$y_p'' = -9A \sin 3x - 9B \cos 3x$$

Hence,

$$\begin{aligned}
& -9A \sin 3x - 9B \cos 3x \\
& + 12A \cos 3x - 12B \sin 3x \\
& + 3A \sin 3x + 3B \cos 3x \\
& = \sin 3x
\end{aligned}$$

$$\begin{aligned}
& (-9A - 12B + 3A) \sin 3x \\
& + (-9B + 12A + 3B) \cos 3x \\
& = \sin 3x
\end{aligned}$$

$$-6A - 12B = 1$$

$$-6B + 12A = 0$$

$$-24B - 6B = 2$$

$$\Rightarrow B = -1/15 \leftarrow$$

$$\Rightarrow -6A = 1 + 12(-1/15)$$

$$-6A = 3/15$$

$$A = -1/30 \leftarrow$$

$$\Rightarrow y_p = \frac{-1}{30} \sin 3x - \frac{1}{15} \cos 3x$$

$$GS = y_p + y_h.$$

$$y'' + 4y' + 3y = x^2$$

$$y_h = C_1 e^{-3x} + C_2 e^{-x}$$

Suppose $ax^2 + bx + c = y_p$
is a sol.

$$y_p' = 2ax + b$$

$$y_p'' = 2a$$

$$\Rightarrow 2a + \cancel{8ax} + 4b + \cancel{3ax} + \cancel{3bx} + 3c = x^2$$

$$\Rightarrow 3a = 1$$

$$8a + 3b = 0$$

$$2a + 4b + 3c = 0$$

$$\Rightarrow a = 1/3$$

$$b = -\frac{8}{3} = -\frac{8}{3}$$

$$3 \cdot 3$$

$$3c = -\frac{2}{3} + \frac{32}{9}$$

$$= \frac{26}{9}$$

$$\Rightarrow c = \frac{26}{27}$$